

Task 1: Access Nested Object Properties

Instructions:

Given the following object, access and log the following properties:

- The first employee's name and age.
- The manager's phone number.
- The first product's price.

```
const company = {
  employees: [
    { name: "Alice", age: 28, position: "Developer" },
    { name: "Bob", age: 35, position: "Manager" }
  ],
  products: [
    { name: "Laptop", price: 999 },
    { name: "Phone", price: 499 }
  ],
  managerContact: {
    phone: "555-1234",
    email: "bob@company.com"
  }
};
```

Task:

- Log **Alice's name** and **age**.
 - Log the **manager's phone number**.
 - Log the **price of the first product**.
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Task 2: Modify Nested Array Elements

Instructions:

You are given the following shopping list. Modify the price of the second item in the `groceryList` array to 4.99 and then log the updated array.

```
const shoppingList = {
  groceries: [
    { name: "Milk", price: 2.49 },
    { name: "Eggs", price: 3.99 },
    { name: "Bread", price: 1.89 }
  ]
};
```

Task:

- Update the **price of Eggs** to 4.99.
- Log the entire `groceries` array after the update.

Task 3: Access Nested Arrays and Objects

Instructions:

Given this object structure for a school's class data, access and log the names of the students in the class.

```
const school = {  
  class: "10th Grade",  
  students: [  
    { name: "John", subjects: ["Math", "Science", "History"] },  
    { name: "Mia", subjects: ["English", "Geography", "Math"] }  
  ]  
};
```

Task:

- Log the names of the **students** in the class.
 - Log the **subjects** of the first student.
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Task 4: Working with Nested Arrays in Objects

Instructions:

You are given the following object. Access the second item in the `skills` array for the first person, and then access the third item in the `skills` array for the second person.

```
const team = {  
  members: [  
    { name: "Jake", skills: ["JavaScript", "React", "Node.js"] },  
    { name: "Sara", skills: ["HTML", "CSS", "JavaScript"] }  
  ]  
};
```

Task:

- Log **Jake's** second skill.
 - Log **Sara's** third skill.
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Task 5: Deeply Nested Access

Instructions:

The following object represents the details of a car. You need to access and log the following:

1. The **year** of the car.
2. The **second feature** of the car.
3. The **color** of the car.

```
const car = {
  make: "Toyota",
  model: "Corolla",
  details: {
    year: 2020,
    features: ["Bluetooth", "Backup Camera", "Sunroof"],
    color: "Blue"
  }
};
```

Task:

- Log the **year** of the car.
 - Log the **second feature** of the car.
 - Log the **color** of the car.
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Task 6: Access Nested Data in an Array of Objects

Instructions:

You are given an array of students with their grades. Access and log the **first grade** of the second student and the **name** of the first student.

```
const students = [
  { name: "Emily", grades: [85, 92, 88] },
  { name: "Liam", grades: [78, 84, 91] }
];
```

Task:

- Log the **name** of the first student.
 - Log the **first grade** of the second student.
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Task 7: Nested Arrays with Objects

Instructions:

You are given a shopping list with items categorized by type. Access and log the price of the first fruit.

```
const shoppingList = {
  fruits: [
    { name: "Apple", price: 1.5 },
```

```
    { name: "Banana", price: 0.9 }
  ],
  vegetables: [
    { name: "Carrot", price: 2.0 },
    { name: "Spinach", price: 1.8 }
  ]
};
```

Task:

- Log the **price of the first fruit** (Apple).

Task 8: Modify Nested Array Values

Instructions:

Given the following `schedule` object, update the time of the second class and log the updated schedule.

```
const schedule = {
  classes: [
    { name: "Math", time: "9:00 AM" },
    { name: "English", time: "11:00 AM" },
    { name: "Science", time: "1:00 PM" }
  ]
};
```

Task:

- Change the **time of the second class** (English) to "10:00 AM".
- Log the entire `classes` array after the update.

Task 9: Deeply Nested Object Access

Instructions:

Given the following structure for a user's profile, access and log the user's **address** (street and city).

```
const userProfile = {
  username: "john_doe",
  profileDetails: {
    name: "John Doe",
    age: 29,
    address: {
      street: "123 Main St",
      city: "Springfield"
    }
  }
};
```

```
    }  
  };
```

Task:

- Log the **street** and **city** of the user's address.
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Task 10: Modify Nested Array Inside Object

Instructions:

You are given the following object, which represents a user's playlist. Add a new song to the `favorites` array and log the updated array.

```
const userPlaylist = {  
  username: "jane_smith",  
  favorites: ["Song 1", "Song 2", "Song 3"]  
};
```

Task:

- Add a new song "Song 4" to the `favorites` array.
- Log the updated `favorites` array.